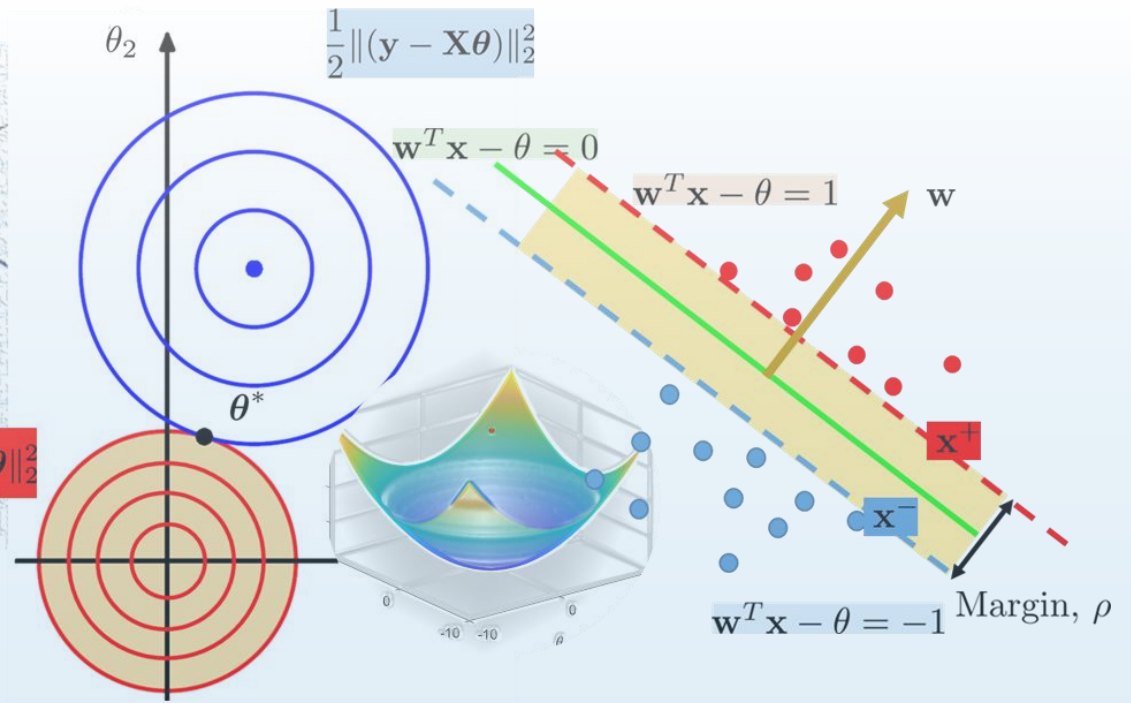
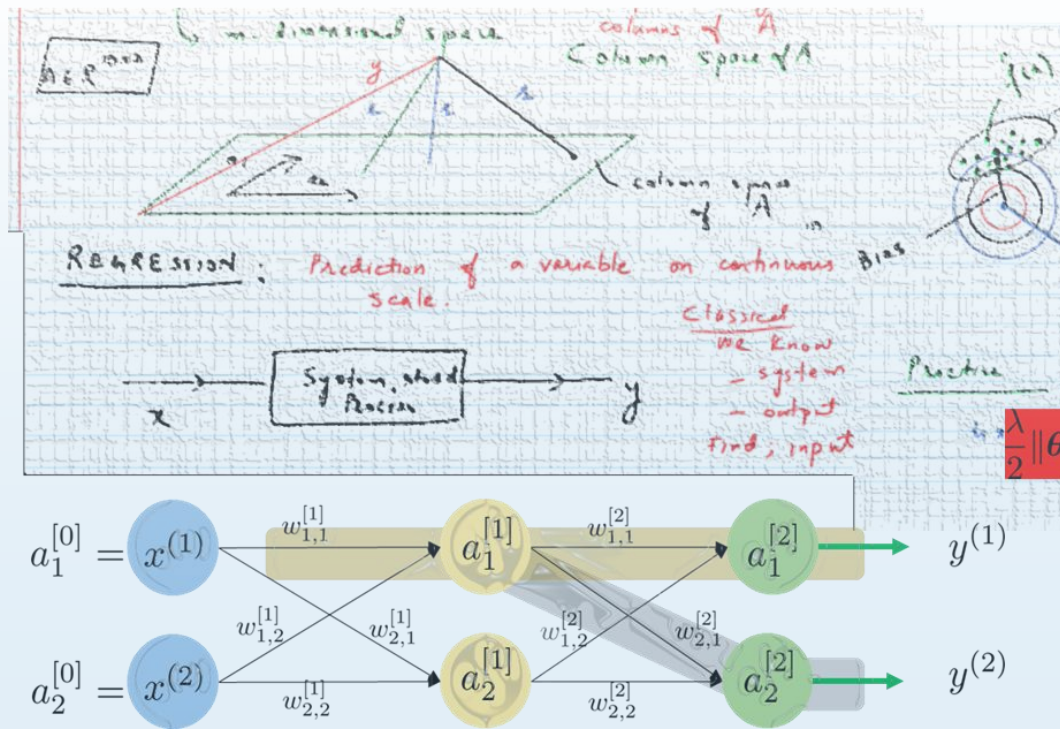




AI-501 Mathematics for AI

Overview

Zubair Khalid
School of Science and Engineering

Outline

- *Course Overview, Logistics*
- *AI Overview*
- *Introduction to Machine Learning*

About us!



Salaar



Zubair



Nausherwan



Afsheen



Haseeb



Shawal



Sabeer



Muhammad

About the Instructor (Zubair Khalid)

- Associate Professor, LUMS
- Post-doctorate – 2013-2015, Australian National University (ANU)
- PhD, Australian National University (ANU) - 2013
- CITY – Centre for Urban Informatics, Technology and Policy (www.city.lums.edu.pk)
- Academic webpage: www.zubairkhalid.org

Collaborations: Princeton, UCL, University of Edinburgh, EPFL, ANU, KAUST

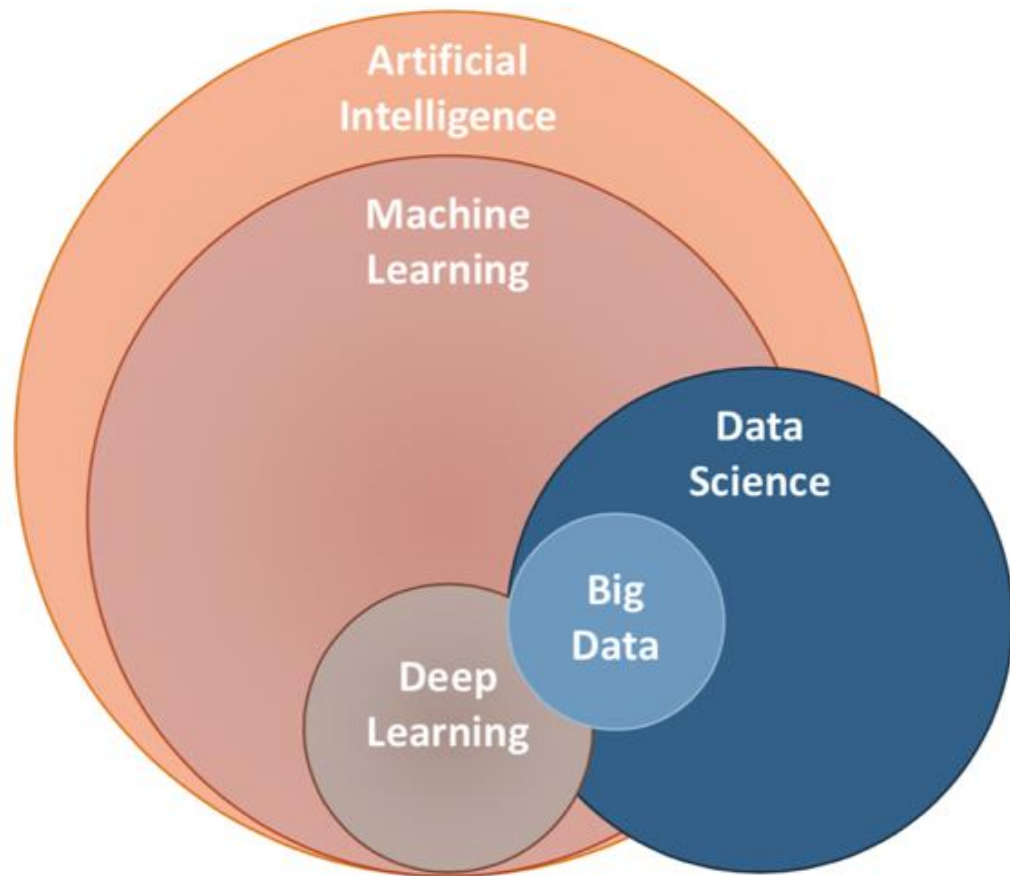
PhD Students: 10 (7 graduated)

Publications: More than 92 (33 Transactions/Journals, 58 Conference proceedings)

Service: Senior Member IEEE and Associate Editor, IEEE Signal Processing Letters

- Gordon-Bell Prize in Climate Modeling, 2024 (Termed as Nobel Prize in Supercomputing)

Overview



Is it possible to learn
artificial intelligence
without mathematical
knowledge?

AI and Machine Learning

- *Data*
 - *visualizations, operations, manipulations*
- *Analysis*
 - *insights, statistics, inference*
- *Model*
 - *representation of process, assumptions*
- *Learning*
 - *Optimize model with respect to data*

Linear Algebra

*Analytic
Geometry*

Calculus

*Probability
Theory*

Course Objectives

- **Foundational Knowledge:** Equip students with a robust understanding of linear algebra, calculus, probability theory, and optimization techniques, which are essential for ML and AI.
- **Practical Applications:** Demonstrate the application of mathematical concepts in real-world ML and AI scenarios, including data processing, model training, and performance evaluation.
- **Analytical Thinking:** Foster analytical thinking by encouraging students to critically assess different algorithms and their suitability for various tasks, considering both theoretical foundations and practical applications.

Is this course a right choice for you?

Graduate students

- *Interesting in pursuing a professional career focused on the development of AI solutions*
- *Interested in pursuing AI, Deep Learning (DL) and Machine Learning (ML) in their grad school (PhD)*
- *Want to do fundamental research in the area of AI*
- *Wish to apply AI in their research work*

Pre-requisites

- *Encouraged to revise Linear Algebra and Probability concepts*

Learning Interface

Communication and Interaction:

- Class:
 - Sunday, 9 am to 12 pm Slot
 - Break: 9:15 to 10:30
 - Quiz in class
 - Engagement in class



Course Page: LMS

Slack: Course-related questions or discussions. We will try to respond to the queries ASAP.

https://join.slack.com/t/ai501mathemat-xwh4968/shared_invite/zt-3cnndxipv-YJQrVE2~VLDTXIWbjEWOZQ

Office Hours: Posted on course page; distributed throughout the week. TAs during office hours will be available online as well.

Grading Distribution and Logistics

- *5 Homeworks: 15%*
- *Quizzes: 15%*
- *Mid-Exam: 30%*
- *Final Exam: 40%*

Course Policies

- *Homework Late Policy*
 - 10% per day for 3 days. No submission after 3 days (72 hours)
- *Missed Quiz Policy*
 - No make-up for quiz
- *Plagiarism will be strictly dealt with as per university policies (take it seriously).*
- *Zero Tolerance for Plagiarism and Cheating*
- *Re-grading can be requested after grade reporting, within the following time limits:*
 - HW and Assignments: 2 days
 - Final Exam: 3 days

Course Policies

Harassment Policy

Harassment of any kind is **unacceptable**, whether it be sexual harassment, online harassment, bullying, coercion, stalking, verbal or physical abuse of any kind. Harassment is a very broad term; it includes both direct and indirect behaviour, it may be physical or psychological in nature, it may be perpetrated online or offline, on campus and off campus. It may be one offense, or it may comprise of several incidents which together amount to sexual harassment. It may include overt requests for sexual favours but can also constitute verbal or written communication of a loaded nature. Further details of what may constitute harassment may be found in the LUMS Sexual Harassment Policy, which is available as part of the university code of conduct.

LUMS has a Sexual Harassment Policy and a Sexual Harassment Inquiry Committee (SHIC). Any member of the LUMS community can file a formal or informal complaint with the SHIC. If you are unsure about the process of filing a complaint, wish to discuss your options or have any questions, concerns, or complaints, please write to the Office of Accessibility and Inclusion (OAI, oi@lums.edu.pk) and SHIC (shic@lums.edu.pk) —both of them exist to help and support you and they will do their best to assist you in whatever way they can.

To file a complaint, please write to harassment@lums.edu.pk.

Course Polices

Help related to equity and Belonging at SSE

SSE's Council on Equity and Belonging is committed to devising ways to provide a safe, inclusive, and respectful learning, living, and working environment for its students, faculty, and staff.

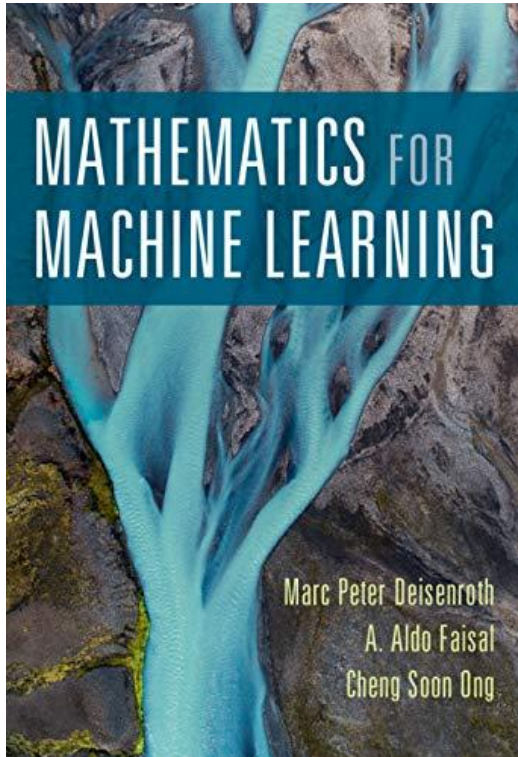
For help related to any such issue, please feel free to write to any member of the school council for help or feedback.

Mental Health Support at LUMS

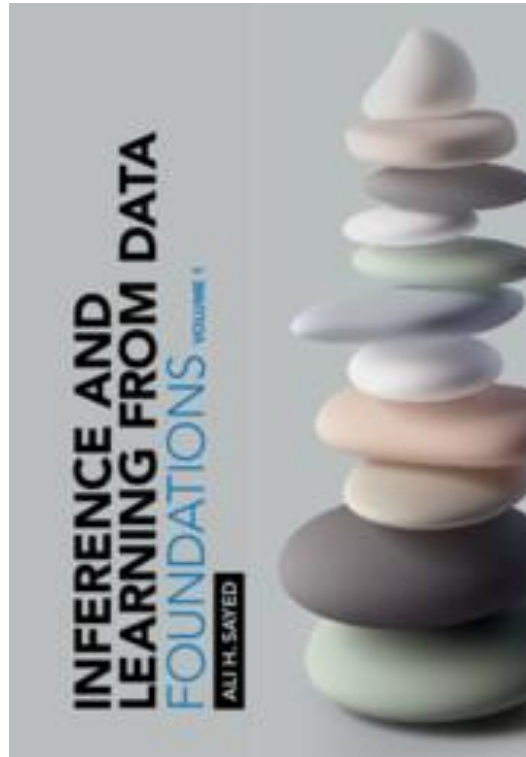
For matters relating to counselling, kindly email student.counselling@lums.edu.pk, or visit <https://osa.lums.edu.pk/content/student-counselling-office> for more information.

You are welcome to write to me or speak to me if you find that your mental health is impacting your ability to participate in the course. However, should you choose not to do so, please contact the Counselling Unit and speak to a counsellor or speak to the OSA team and ask them to write to me so that any necessary accommodations can be made.

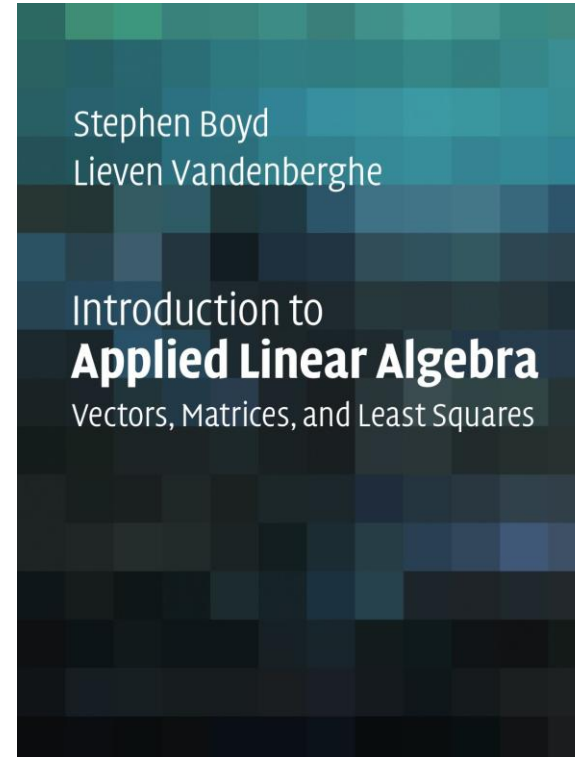
Suggested Reference Books



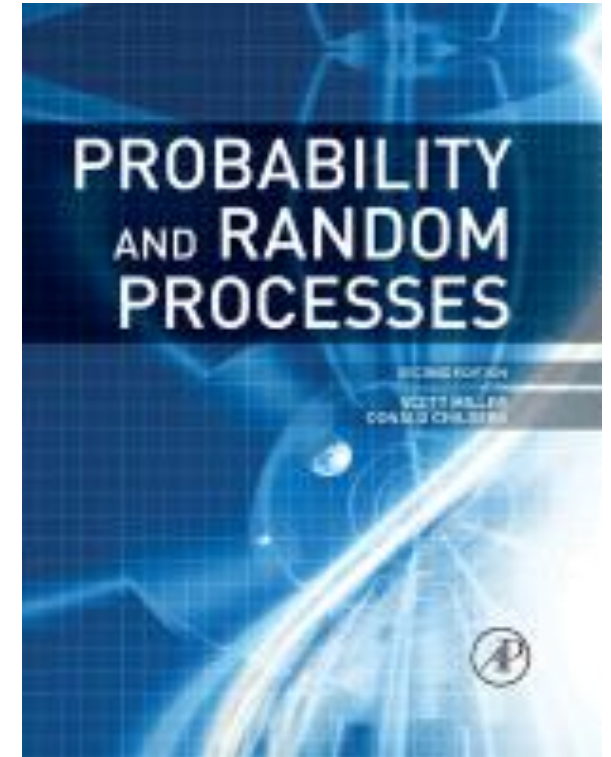
Primary



Reference



Reference



Reference

- *Lecture Notes/Slides will be shared.*

"As to methods, there may be a million and then some, but principles are few. The man who grasps principles can successfully select his own methods."

Ralph Waldo Emerson

Outline

- *Course Overview, Logistics*
- *AI Overview*
- *Introduction to Machine Learning*

AI Overview

What is Intelligence?

“Intelligence is the ability to learn, understand, and apply **knowledge** to **adapt** to **new** situations, reason, and solve problems.”

- Human Intelligence
 - Problem-solving and reasoning
 - Learning from Experience
 - Adaptability
 - Emotional Intelligence

AI Overview

What is Artificial Intelligence (AI)?

“AI refers to the simulation of **human intelligence** in machines that are programmed to think and learn like humans.”

- Narrow AI vs General AI
- SuperIntelligence



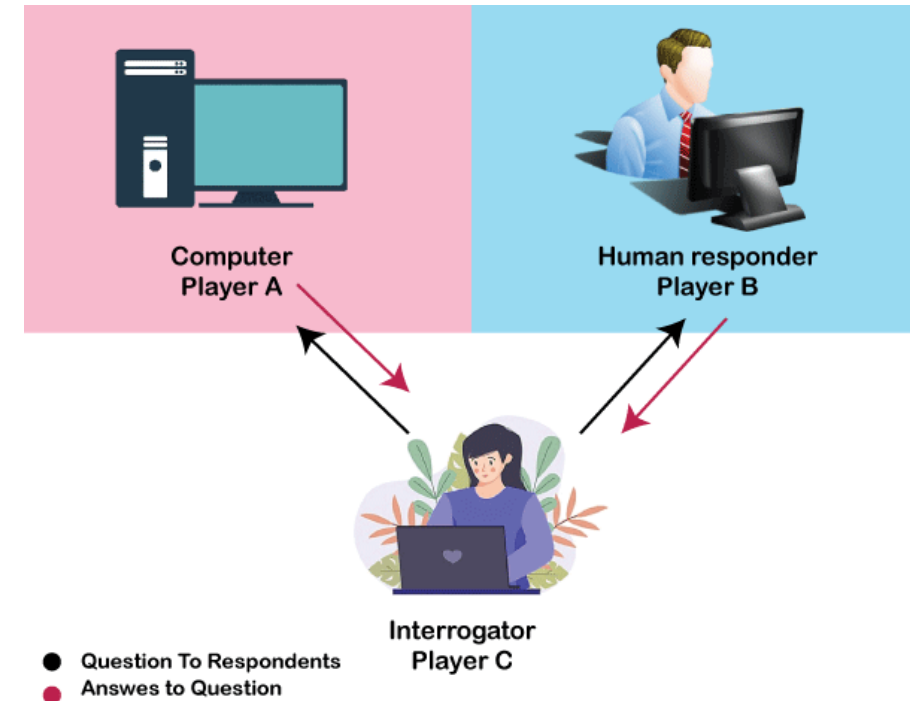
The term AI was coined in 1956 by John McCarthy at a conference.

AI Overview

History of AI – Turing Test

Alan Turing (1950) "Computing machinery and intelligence"

- Can machines think or behave intelligently?
- The Turing Test (imitation game) was designed as a way to judge the success or otherwise of an attempt to produce a thinking computer.
- Predicted that by 2000, a machine would have a 30% chance of fooling a person for 5 minutes
- What do you think?



AI Overview

History of AI – 1960s to 1990s

- Since the 1950s, much of the initial optimism surrounding AI has given way to a more realistic perspective.
- Key areas of focus include:
 - Machine learning
 - Multi-agent systems
 - Computer vision
 - Planning
 - Game Theory
 - Optimization, ...
- 1997 (IBM's Deep Blue), 2012 (Deep Learning breakthrough).

Applications of AI

AI in Healthcare

- *AI-powered Diagnostics*
- *AI systems can analyze medical images, such as X-rays, MRIs, and CT scans, to detect diseases like cancer, heart conditions, and more.*
- *These systems can identify patterns that might be missed by human eyes, resulting in earlier and more accurate diagnoses.*
- ***Example:** Google's DeepMind developed an AI system capable of diagnosing eye diseases with an accuracy that matches world-leading doctors.*

Applications of AI

AI in Healthcare

- *Predictive Analytics*
- *AI can analyze patient data and predict potential health risks, such as the likelihood of developing chronic diseases like diabetes or heart disease. This helps doctors create personalized treatment plans and intervene early.*
- *Example: IBM Watson Health analyzes vast datasets to predict disease progression and recommend personalized treatment options.*

Applications of AI

AI in Healthcare

- *Robotic Surgery*
- *AI-powered robotic systems assist in precision surgeries, allowing for less invasive procedures, faster recovery times, and reduced human error.*
- *Example: The da Vinci Surgical System uses AI to enhance the precision and control of surgeons during minimally invasive procedures.*

Applications of AI

AI in Healthcare

- *Drug Discovery and Development*
- *AI accelerates the drug discovery process by analyzing complex biological data to identify new drug candidates, significantly reducing the time and cost involved in developing new treatments.*
- ***Example:** AI was used by the pharmaceutical company Insilico Medicine to identify potential new compounds for treating fibrosis, which traditionally would have taken years.*

Applications of AI

AI in Healthcare

- *Virtual Health Assistants*
- *AI chatbots and virtual assistants provide personalized health advice, answer questions, and help patients manage chronic conditions by monitoring symptoms and offering medication reminders.*
- *Example: AI-driven platform allows patients to receive medical advice through their smartphones based on the symptoms they input.*

Applications of AI

AI in Finance

- *Fraud Detection*
- *Algorithmic Trading*
- *Risk Management*
- *Personalized Banking*
- *Credit Scoring and Loan Decisions*
- *Financial Forecasting*

Applications of AI

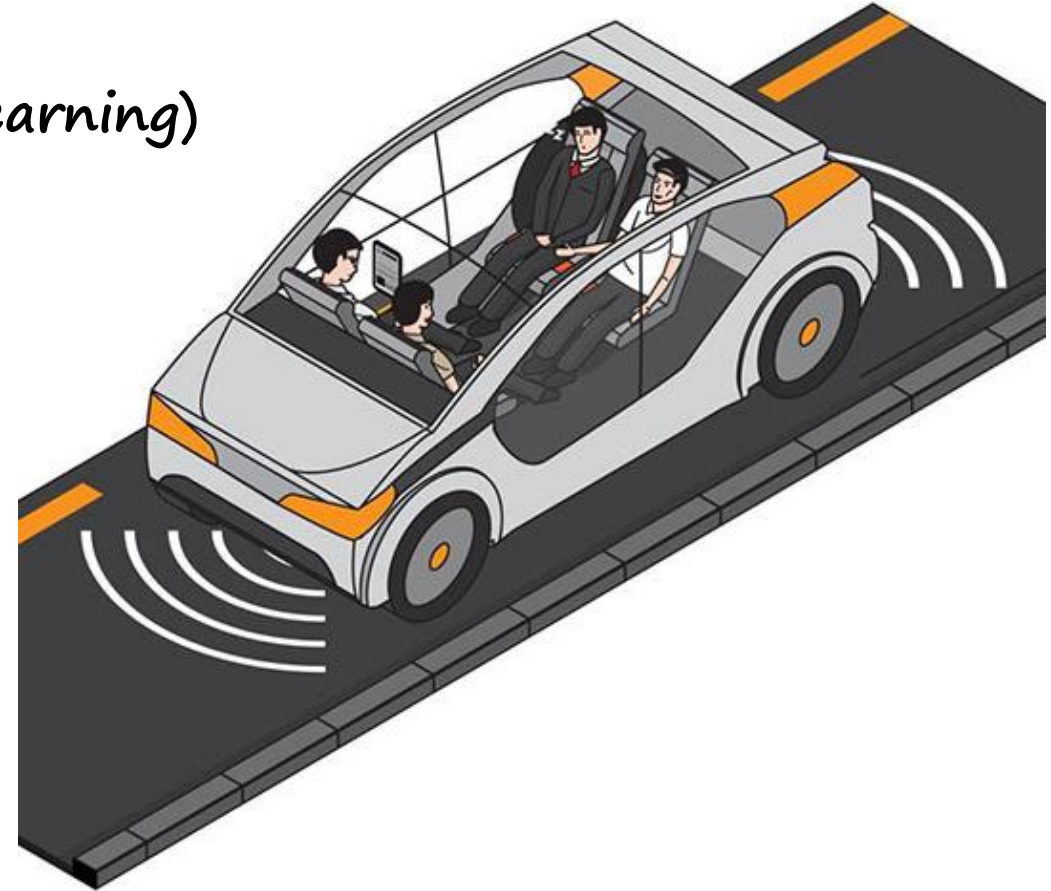
AI in Education

- *Personalized Learning Systems*
- *Virtual Tutors and Chatbots*
- *AI in Assessments and Grading*
- *AI in Special Education*

Applications of AI

AI in Autonomous Vehicles

- *Deep Learning for Object Detection*
- *Decision Making and Planning (Reinforcement Learning)*
- *Simultaneous Localization and Mapping (SLAM)*
- *Path Planning and Navigation*
- *To name a few...*



Applications of AI

AI in Retail

- *Personalized product recommendations*
- *AI-powered chatbots and virtual assistants*
- *Inventory management and demand forecasting*
- *Dynamic pricing*
- *Visual search and image recognition*
- *Customer sentiment analysis*

Applications of AI

AI in Agriculture

- *Precision farming*
- *Crop monitoring and disease detection*
- *Soil health monitoring and analysis*
- *Automated machinery and robotics*
- *Predictive analytics and yield forecasting*

Applications of AI

AI in NLP

- *Virtual assistants (e.g., Siri, Alexa)*
- *Language translation*
- *Text generation and summarization*
- *Speech recognition and transcription*
- *Sentiment analysis*
- *Chatbots for customer support*

Applications of AI

AI in Manufacturing

- *Predictive maintenance*
- *Quality control and defect detection*
- *Supply chain optimization*
- *Autonomous robots and cobots (collaborative robots)*
- *Inventory management and optimization*
- *Production scheduling and process optimization*

Applications of AI

AI in Urban Development

- *Traffic management and optimization*
- *Smart energy grids and resource management*
- *Road safety and surveillance*
- *Waste management optimization*
- *Smart infrastructure and building management*
- *Environmental monitoring*

Centre for Urban Informatics, Technology and Policy (CITY)

Saving our cities through technology and data-driven policy

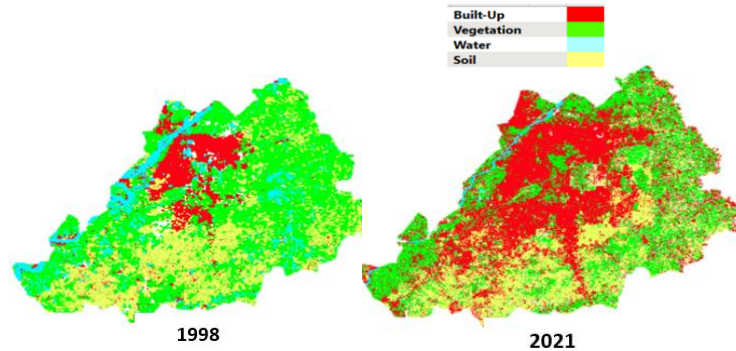
VISION STATEMENT

To become an internationally recognized regional hub of innovation in urban informatics, technology and data-driven policy-making for sustainable urban development.

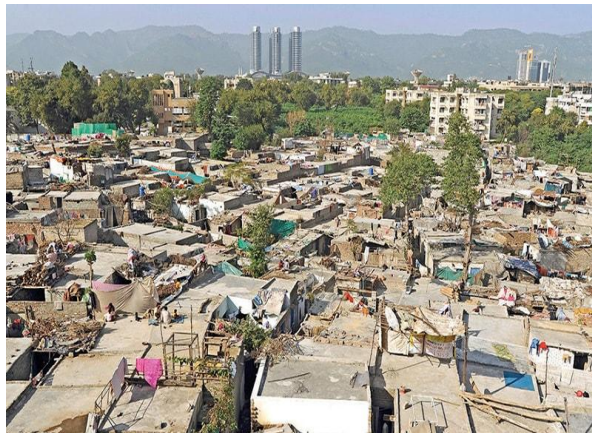


Centre for Urban Informatics, Technology and Policy (CITY)

Growth

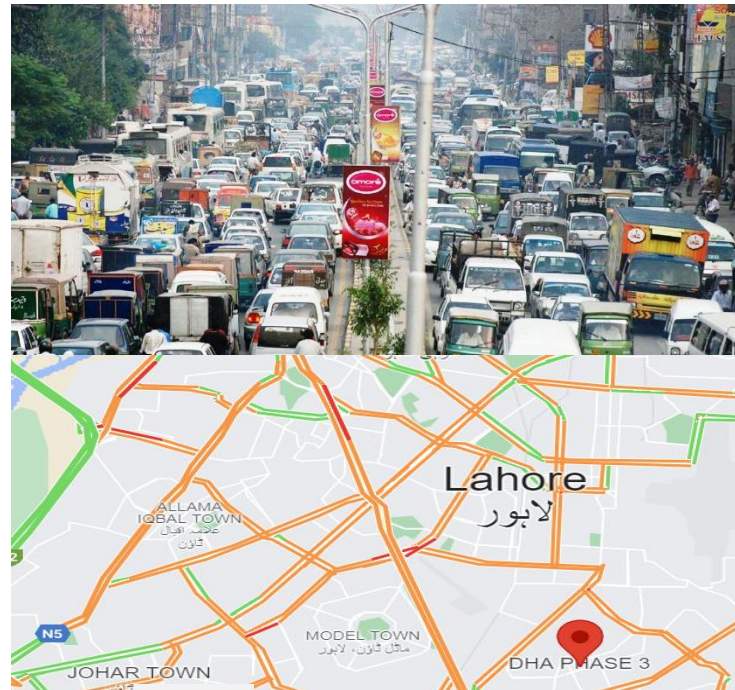


Lahore's Built-up Area. 4X (1998 – 2021)



Murtaza Haider, "Pakistan's urban policy turning cities into slums", Dawn, April 2015.

Mobility



3x (2010-2020)

2x (2010-2020)

Environment and Health

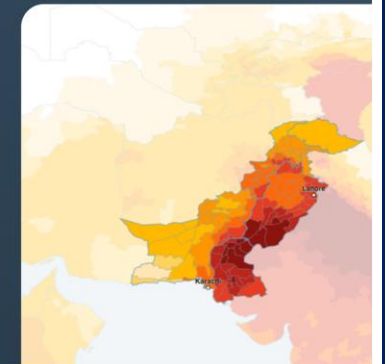


Pakistan

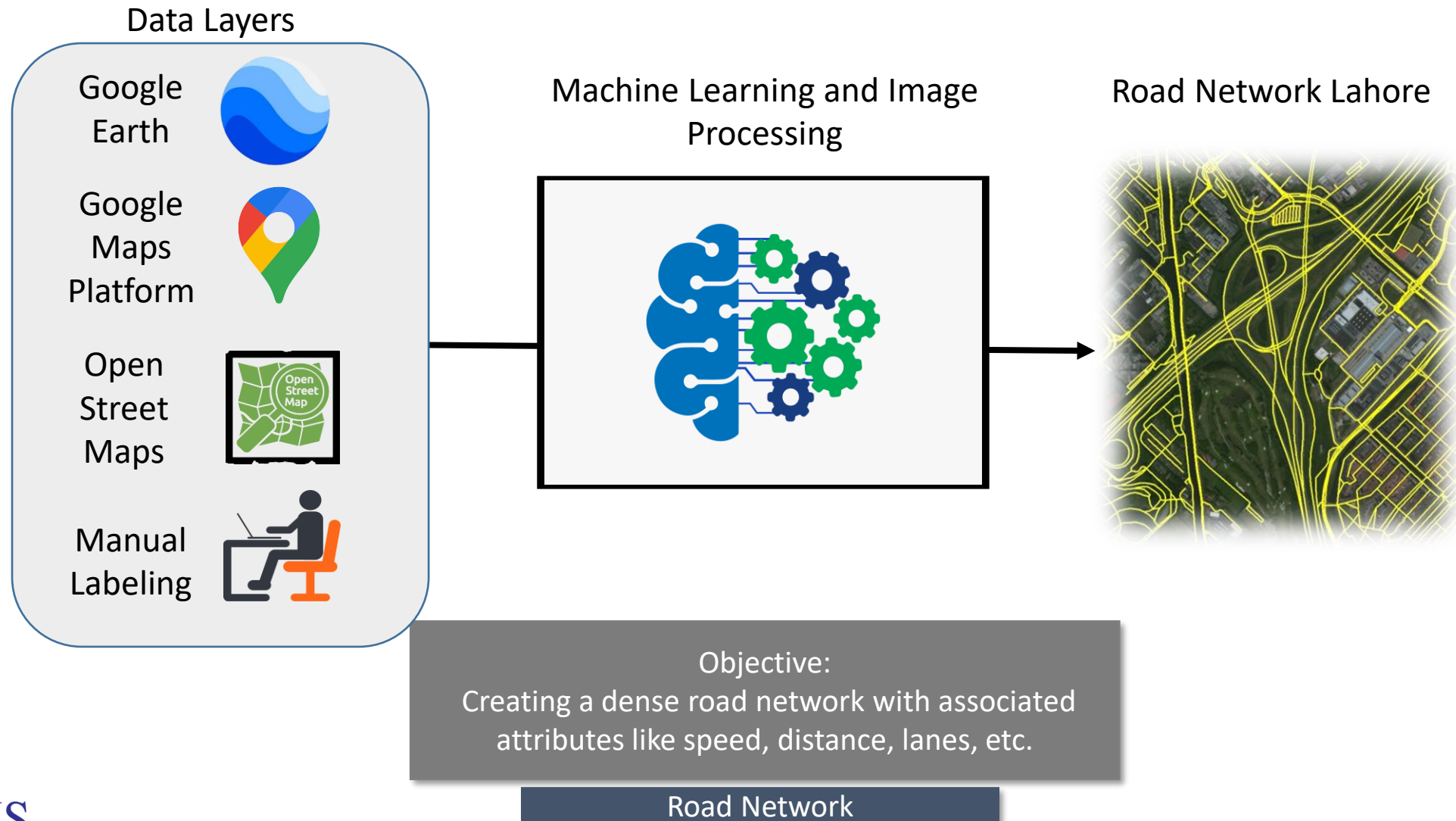
Pollution Ranking
4 out of 243 countries

49.5
Particulate Pollution ($\mu\text{g}/\text{m}^3$)

3.87
Gain in life expectancy if WHO Guideline is met
WHO Guideline: $10 \mu\text{g}/\text{m}^3$



Urban Mobility: Road Network



Urban Mobility: Road Network

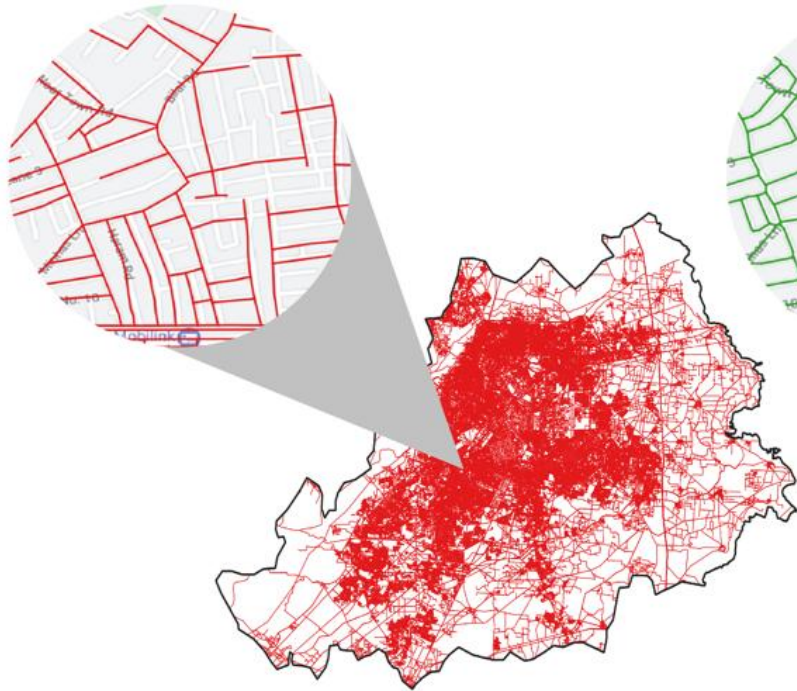


Analysis Tools:

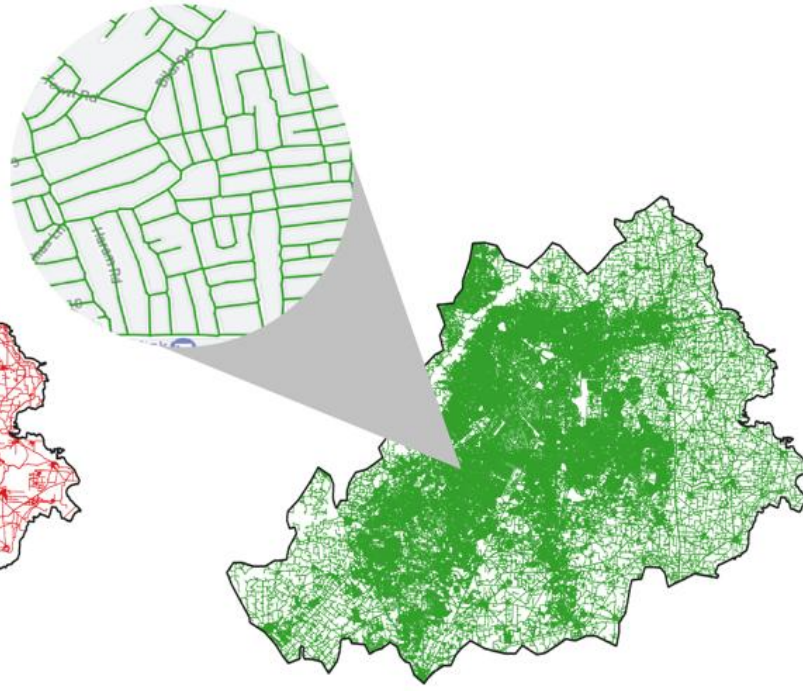
QGIS, Python, OpenCV, and SciPy

Road Network

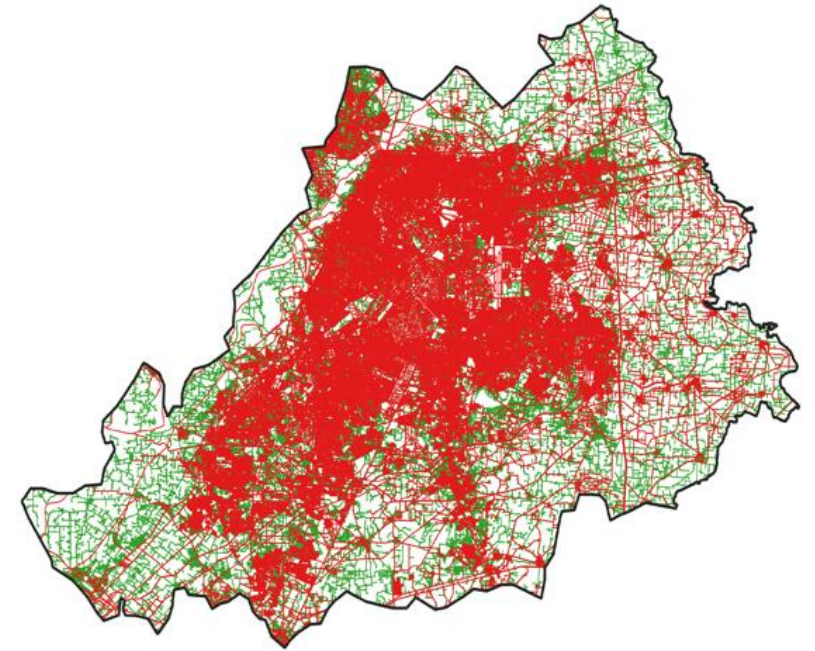
Urban Mobility: Road Network



OSM Road Network

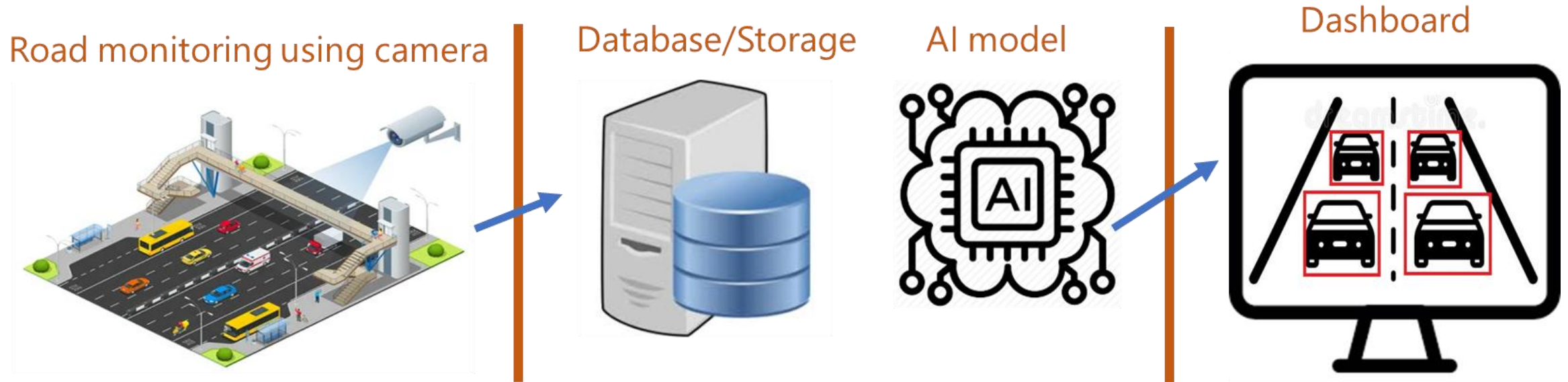


Our Road Network



Overlaid

Urban Mobility: Traffic Informatics



AI based system for real time traffic flow density

Traffic Flows Density

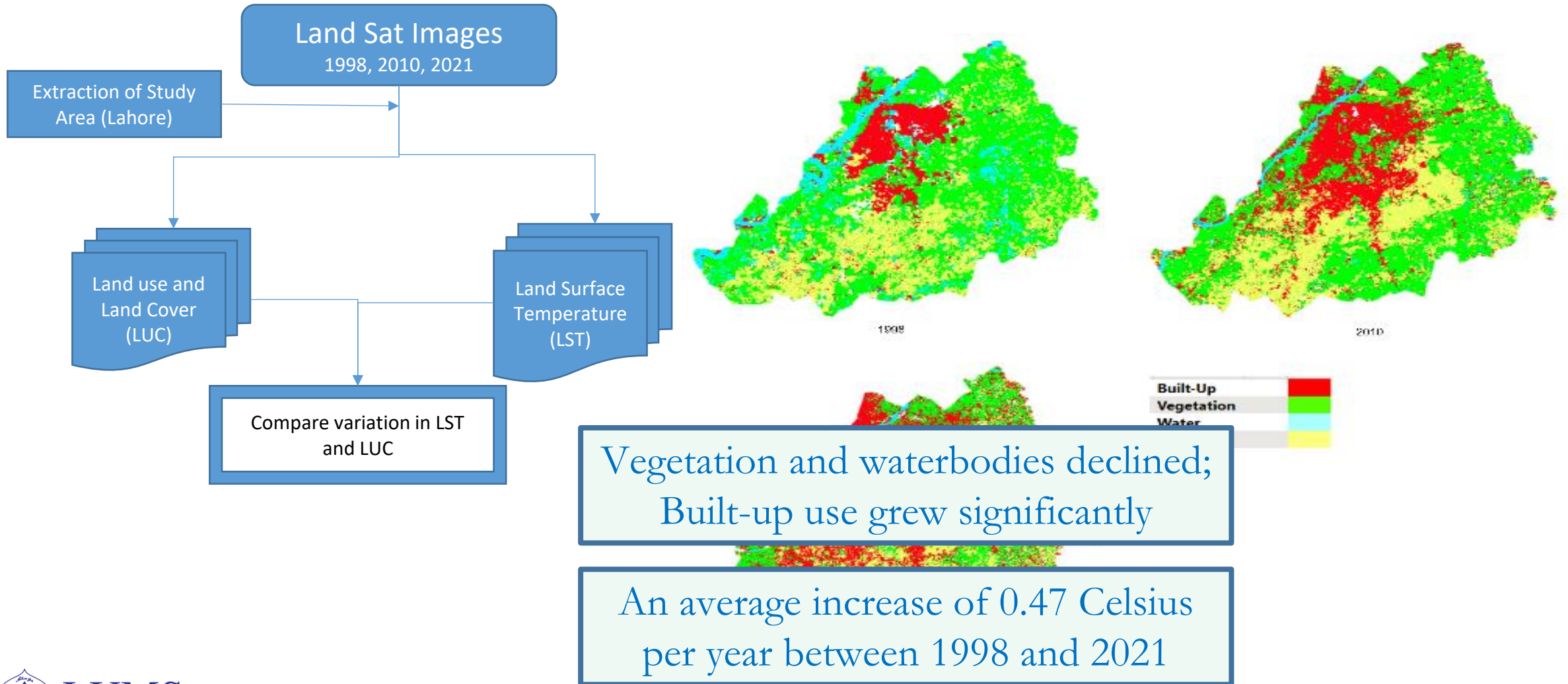
Urban Mobility: Traffic Informatics



Prototype Sample

Traffic Flows Density

Urban Sprawl: Land Surface Temperature



Data-driven Urban Forestation: Tree Cover Estimation

Problem: **Estimate green spaces and tree cover in urban areas**

Why?

Green spaces – significant contributors to ecosystem services: air purification and carbon sequestration

Challenges

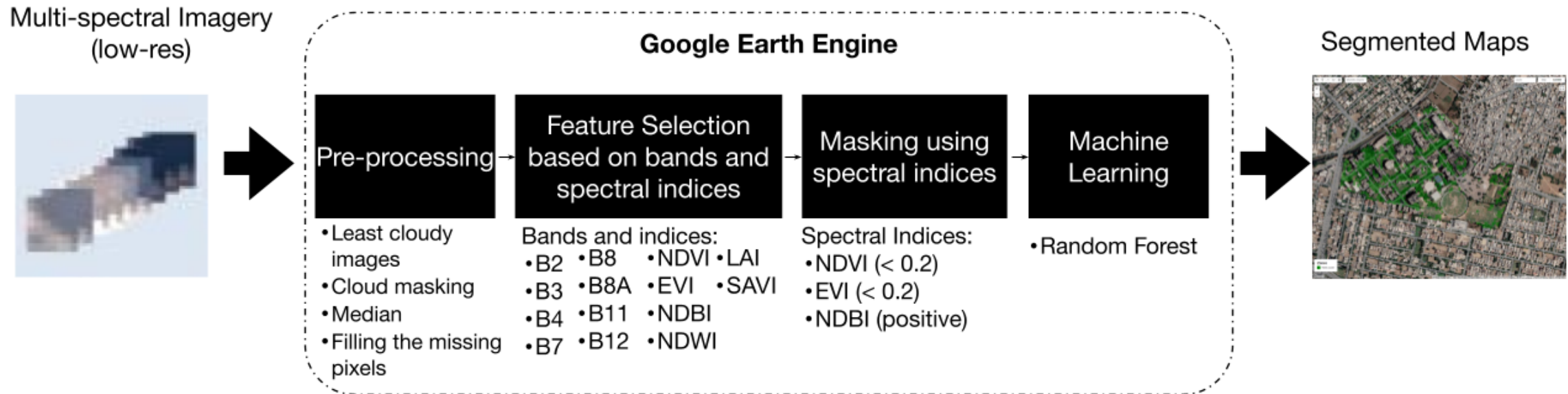
- Bottleneck: absence of spatially explicit data
- Huge variations in satellite imagery

Existing Methodologies

- Current methodologies focus on forest mapping
- A gap exists in urban tree cover estimation in developing countries with low resolution imagery

Data-driven Urban Forestation: Tree Cover Estimation

Tree Cover Estimation using Machine Learning and Feature Selection on Sentinel-2 Data



U. Nazir, M. Uppal, M. Tahir, and Z. Khalid, "Feature Selection on Sentinel-2 Multi-Spectral Imagery for Efficient Tree Cover Estimation," 2023 IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2023), Pasadena, California

Data-driven Urban Forestation: Tree Cover Estimation



LUMS Ground Truth



Proposed



ESA WorldCover

Model	Pred. area (acres)	Masking	Spectral indices	Pixel-wise Test Accuracy (%)	Kappa Score
RF-spectral-bands	29.5	No	No	0.93	0.81
RF-spectral-indices	28	No	Yes	0.95	0.88
Proposed	25	Yes	Yes	0.99	0.92
ESA WorldCover Product ⁵	16	-	-	0.74	-
DeepLabv3 ⁶	28	No	No	0.80	-

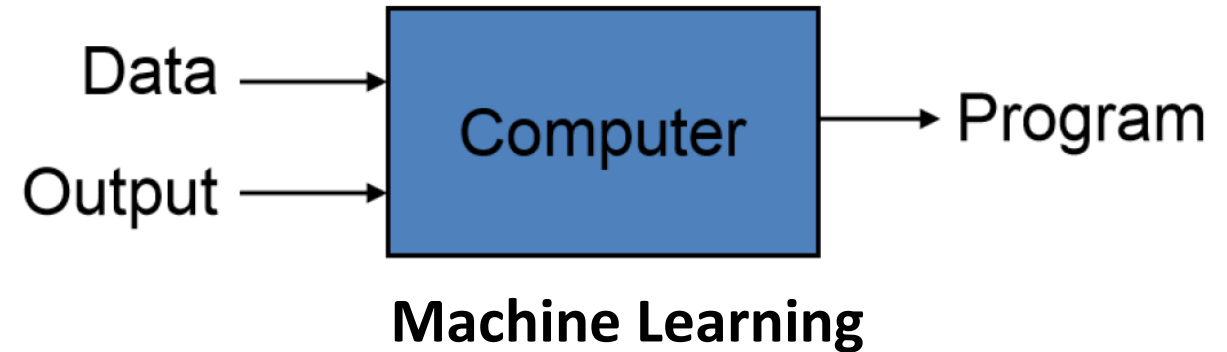
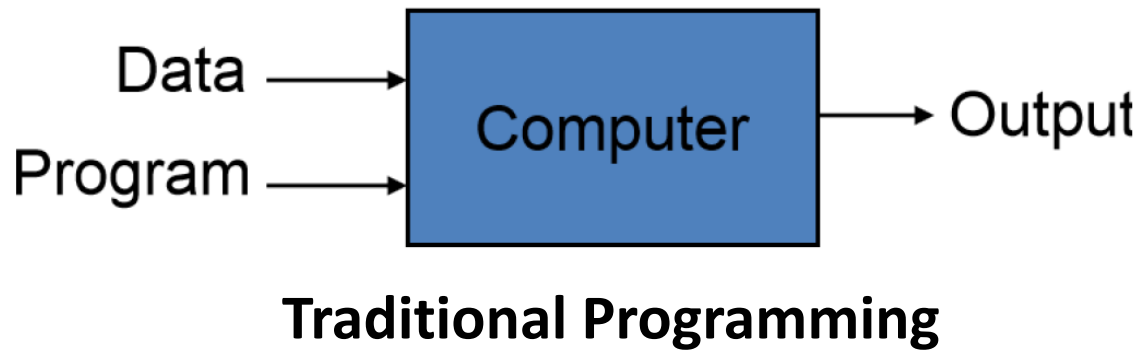
Outline

- *Course Overview, Logistics*
- *AI Overview*
- *Introduction to Machine Learning*

Machine Learning Overview

What is Machine Learning?

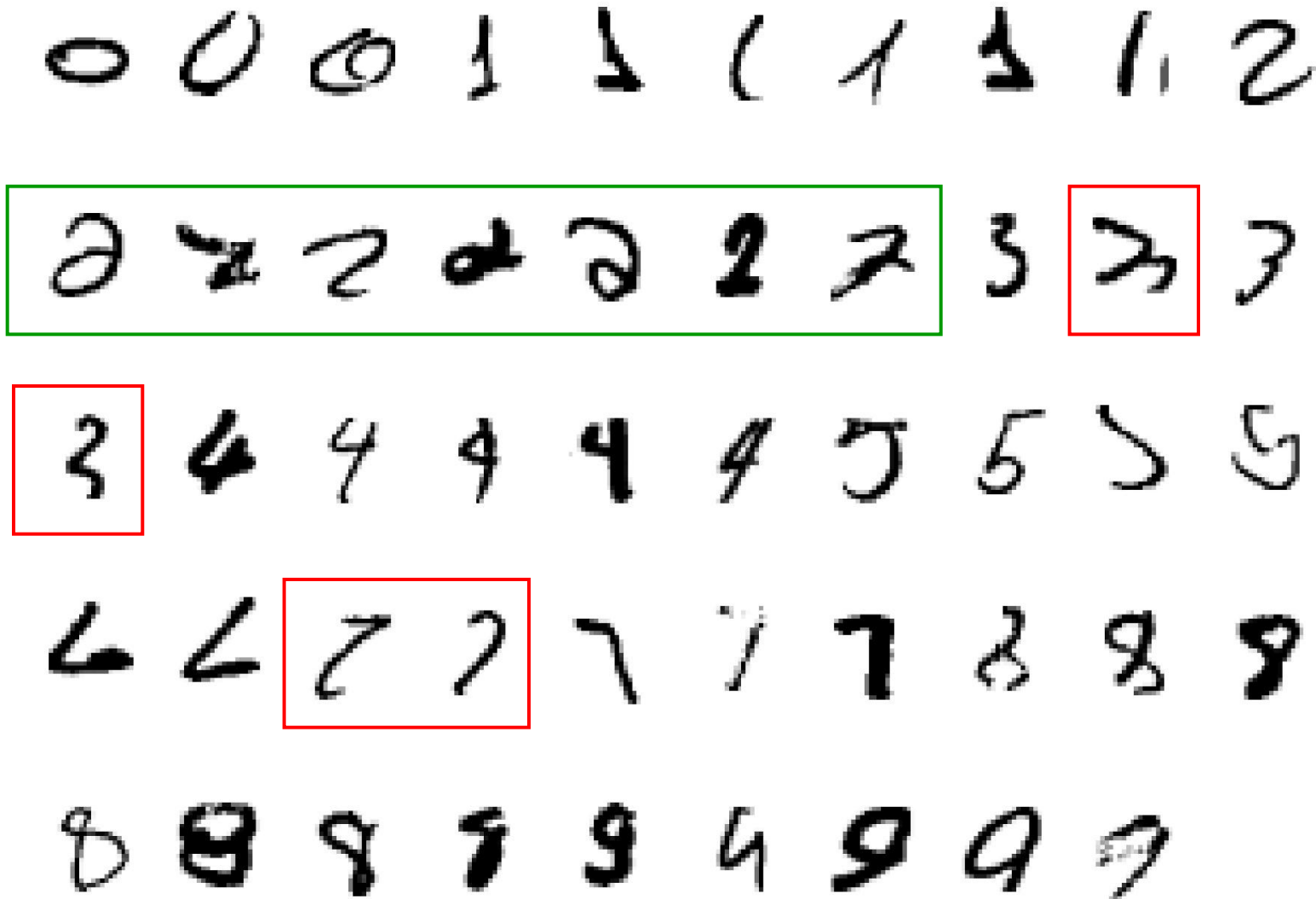
- Automating the process of automation
- Getting computers to program themselves



Given examples (training data), make a machine learn system behavior or discover patterns

Machine Learning Overview

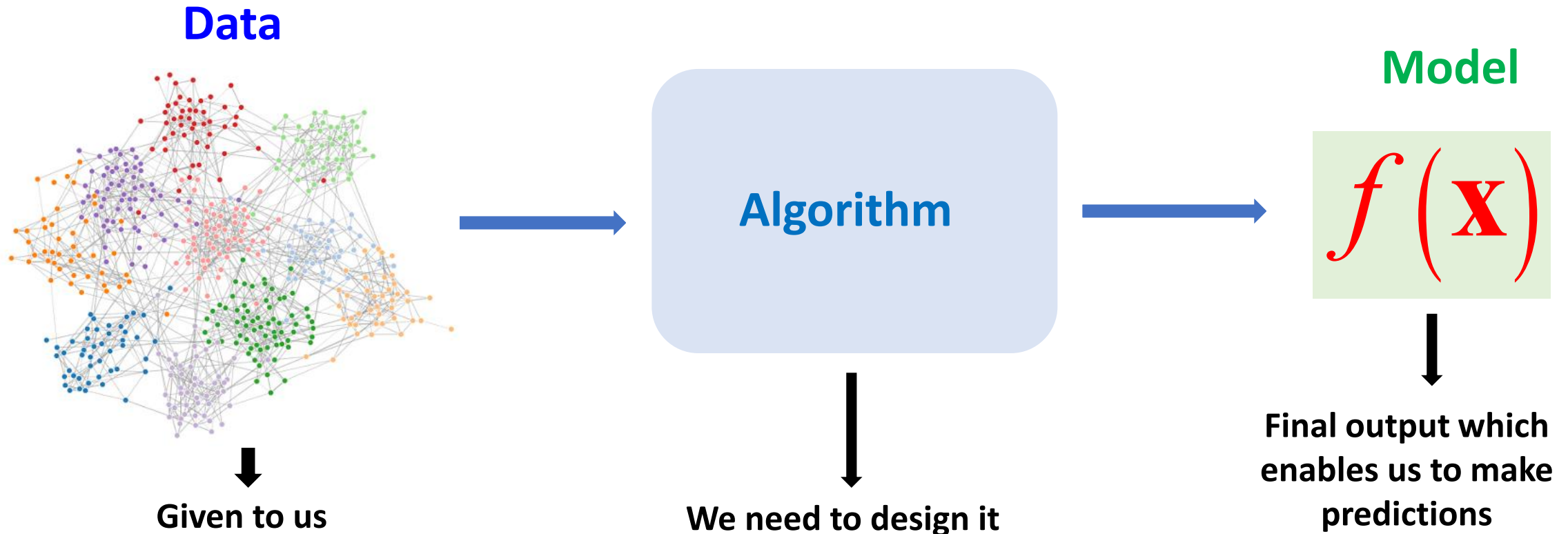
Classical Example: Recognize hand-written 2!



Machine Learning: Overview

What is Machine Learning?

Given examples (training data), make a machine learn system behavior or discover patterns



Machine Learning: Overview

Algorithms vs Model

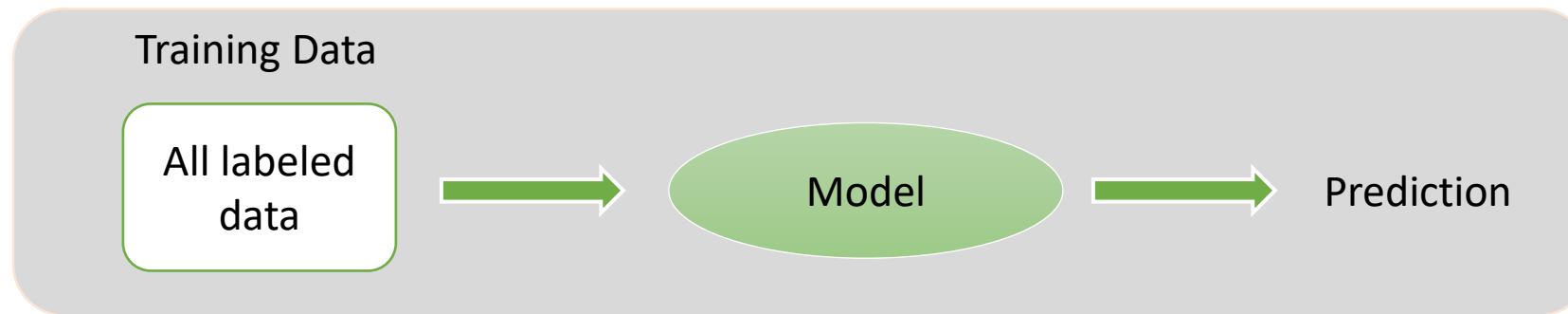
- Linear regression algorithm produces a model, that is, a vector of values of the coefficients of the model.
- Decision tree algorithm produces a model comprised of a tree of if-then statements with specific values.
- Neural network along with backpropagation + gradient descent: produces a model comprised of a trained (weights assigned) neural network.

Machine Learning: Overview

Nature of ML Problems

1. Supervised Learning

The learning algorithm would receive a set of inputs along with the corresponding correct outputs to train a model



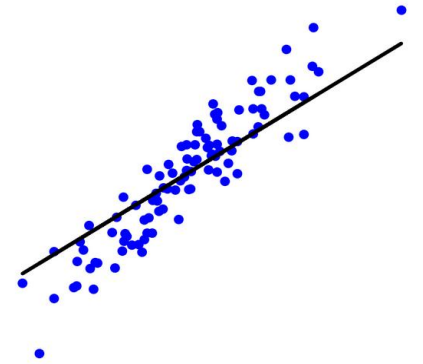
Supervised Learning

Regression

Regression: Quantitative Prediction on a continuous scale

Examples: Prediction of

- Age of a person from his/her photo
- Price of 10 Marla, 5-bedroom house in 2050
- USD/PKR exchange rate after one year
- Efficacy of vaccine or medicine
- Average temperature/Rainfall during monsoon
- Cumulative score in AI501 course
- Probability of decrease in the electricity prices in Pakistan
- No. of steps per day



What do all these problems have in common?

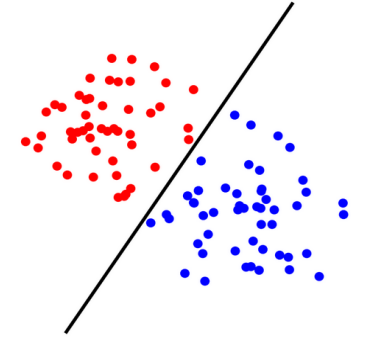
Continuous outputs

Predicting continuous outputs is called regression

Supervised Learning

Classification

Classification: Given a data sample, predict its class (discrete)



Examples: Prediction of

- Gender of a person using his/her photo or hand-writing style
- Spam filtering
- Object or face detection in a photo
- Temperature/Rainfall normal or abnormal during monsoon
- Letter grade in AI501 course
- Decrease expected in electricity prices in Pakistan next year
- More than 10000 Steps taken today

What do all these problems have in common?

Discrete outputs: Categorical

Yes/No (Binary Classification)

Multi-class classification:
multiple classes

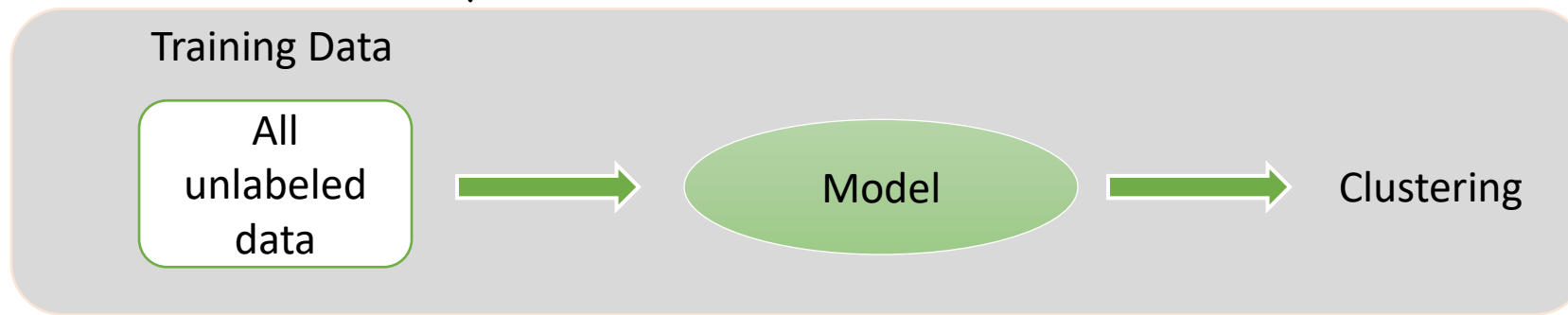
Predicting a categorical output is called classification

Machine Learning: Overview

Nature of ML Problems

2. Unsupervised Learning

The learning algorithm would receive unlabeled raw data to train a model and to find patterns in the data

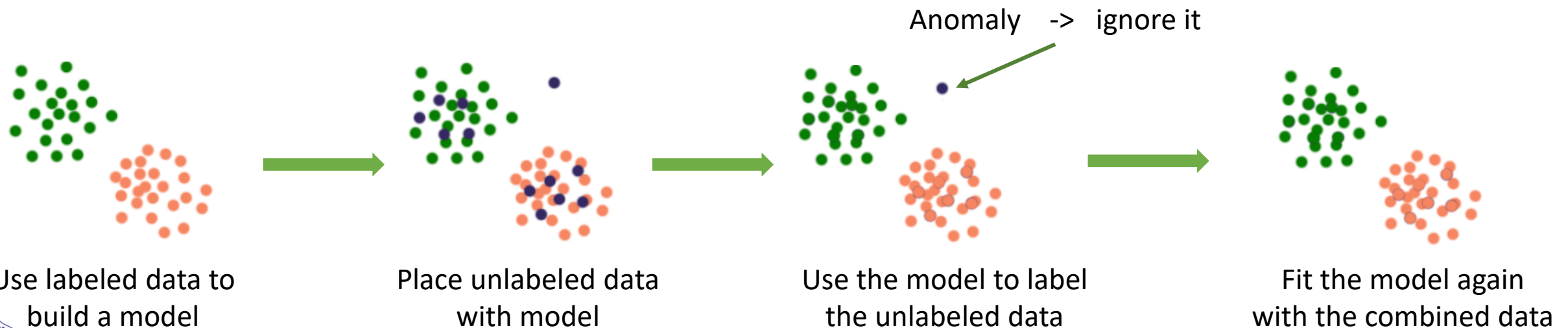
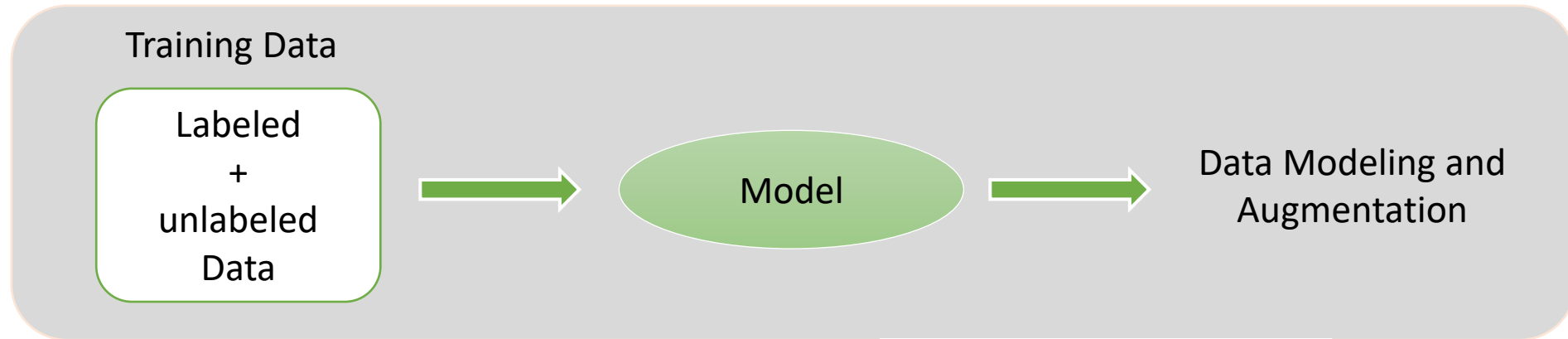


Machine Learning: Overview

Nature of ML Problems

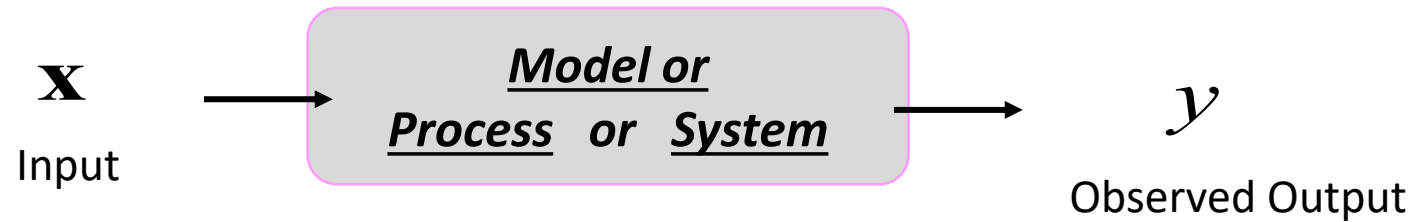
3. Semi-supervised Learning

- The learning algorithm receives labeled and unlabeled raw data to train a model
- Main objective is to efficiently accommodate the unlabeled data



Machine Learning: Overview

Training Data Collection

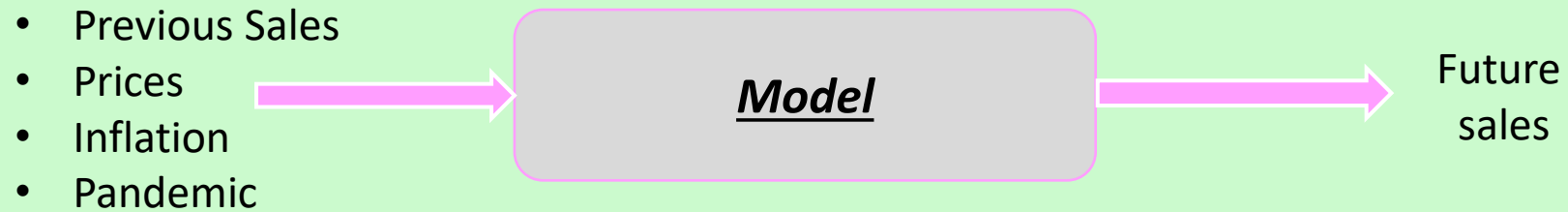


PROCESS or **SYSTEM** : Underlying physical or logical phenomenon which maps our input data to our observed output

Collect the training data by observing our unknown **PROCESS** or **SYSTEM**

Machine Learning: Overview

Example Systems

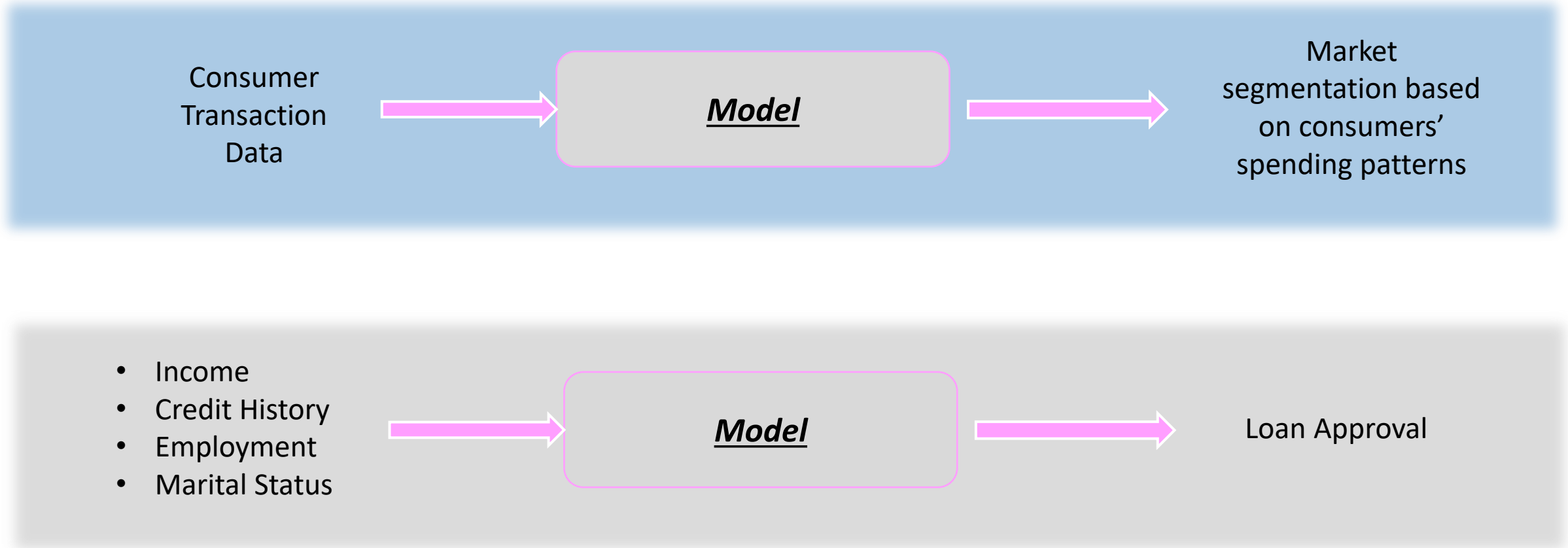


Image



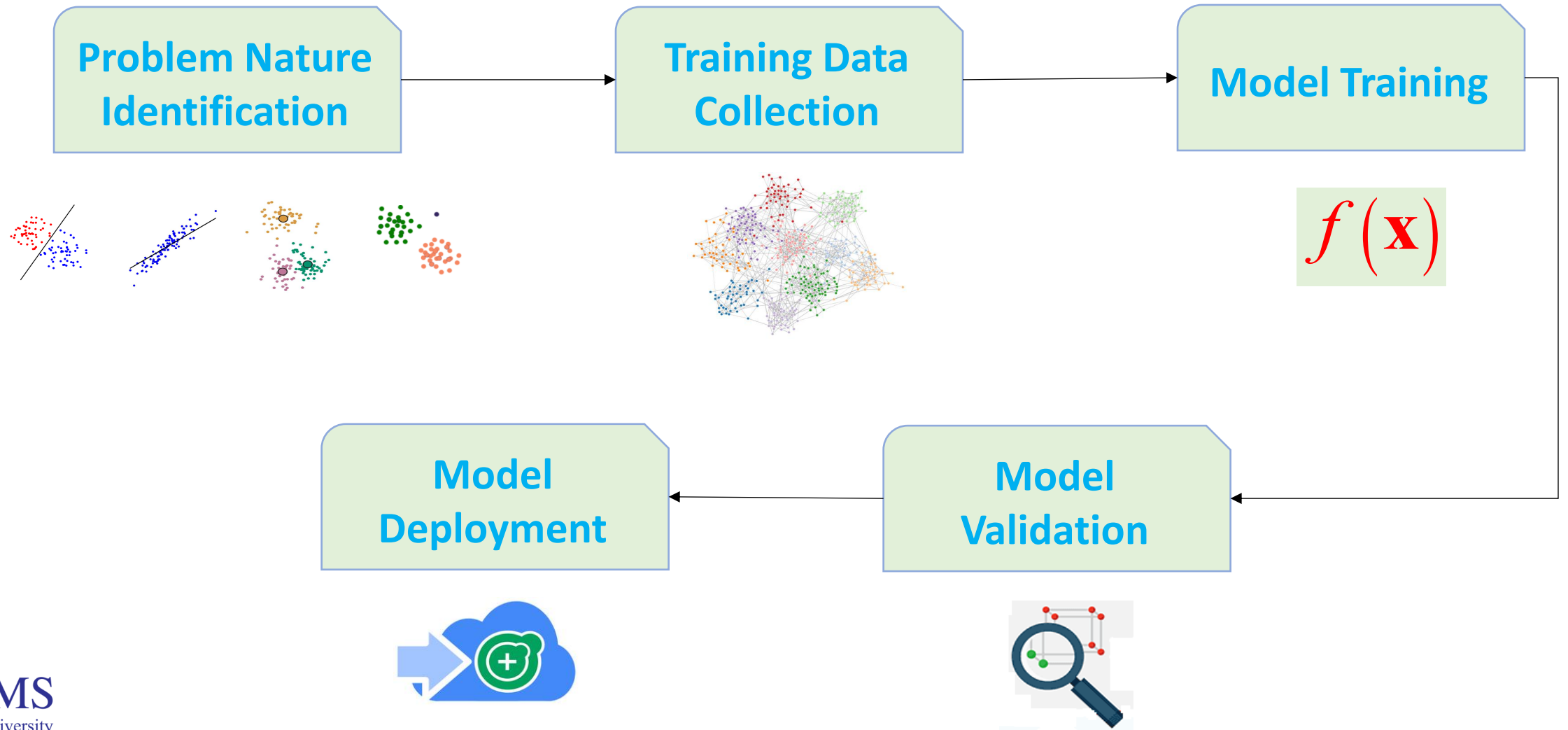
Machine Learning: Overview

Example Systems



Machine Learning: Overview

Typical Flow



Supervised Learning Setup

Nomenclature

In these regression or classification problems, we have

- *Inputs* – referred to as Features
- *Output* – referred to as Label
- *Training data* – (input, output) for which the output is known and is used for training a model by ML algorithm
- *A Loss, an objective or a cost function* – determines how well a trained model approximates the training data
- *Test data* – (input, output) for which the output is known and is used for the evaluation of the performance of the trained model

Supervised Learning Setup

Nomenclature - Example

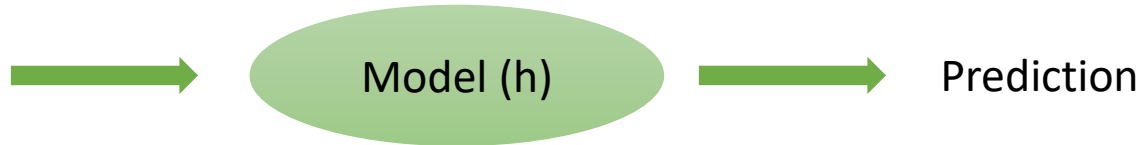
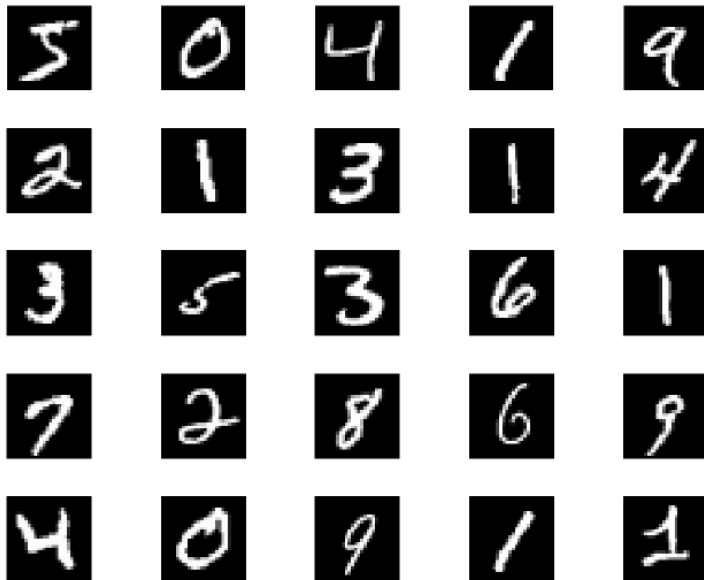
Predict Stock Index Price

- Features (Input)
- Labels (Output)
- Training data

Interest_Rate	Unemployment_Rate	Stock_Index_Price
2.75	5.3	1464
2.5	5.3	1394
2.5	5.3	1357
2.5	5.3	1293
2.5	5.4	1256
2.5	5.6	1254
2.5	5.5	1234
2.25	5.5	1195
2.25	5.5	1159
2.25	5.6	1167
2	5.7	1130
2	5.9	1075
2	6	1047
1.75	5.9	965
1.75	5.8	943
1.75	6.1	958
1.75	6.2	971
1.75	6.1	949
1.75	6.1	884
1.75	6.1	866
1.75	5.9	876
1.75	6.2	?
1.75	6.2	?
1.75	6.1	?

Supervised Learning Setup

Example



MNIST Data:

- Each sample 28x28 pixel image
- 60,000 training data
- 10,000 testing data

Feedback